

Cloud-Based Systems Analysis for Christian Volunteer Service (CVS)

1. SWOT Analysis

1.1 Current IT Infrastructure & Services

Figure: SWOT Analysis — Current IT Infrastructure & Services



Figure 1: SWOT Analysis - Current IT Infrastructure & Services

Strengths	CVS maintains established IT systems holding critical membership and donor records totalling 6.4GB, supported by a dedicated 40-person IT team at head office. The existing CRM, HRM, and payroll systems, despite their age, continue to deliver core operational functionality across the organisation. The current infrastructure provides CVS with full ownership and direct control
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	over sensitive donor and beneficiary data.
Weaknesses	The legacy systems are slow, unreliable, and costly. Last year alone, 15 working days were lost to system downtime, impacting approximately 100 staff per incident and equating to 1,500 lost working days. Annual software licensing and service costs consume £8.5 million, while legacy CRM and database systems specifically cost £7 million per annum. The fragmented IT landscape, with five different payroll packages across 16 regional offices and reliance on Access databases and Excel spreadsheets, creates significant inefficiency and data inconsistency. The multiple IT landscape which has different payroll packages across all the offices create significant inefficiency and data inconsistency.
Opportunities	Merging multiple system could return substantial cost saving and operational efficiency. The integrated system of 6.4 GB of donor and member data could help making data-driven fundraising strategies. There is significant market which are untapped source of revenue for e-commerce for selling the valuable donated items.
Threats	Continued reliance on ageing systems risks catastrophic data loss or extended outages. Rising maintenance costs for legacy systems will progressively consume more of CVS's limited budget. Staff resistance, expressed through union channels, could delay necessary modernisation if not managed through effective change management.

1.2 Proposed IT Infrastructure & Services

Figure: SWOT Analysis — Proposed IT Infrastructure & Services



Figure 3: SWOT Analysis - Proposed IT Infrastructure & Services

Strengths	Saas solutions such as CRM, HRM, and payroll which are cloud based would help reducing the overall licensing costs which also provide automatic updates and improves reliability. The staff of about 4000 would be able to access the system from anywhere. The new revenue sources would be tapped through the centralised e-commerce portal. The shift from Capital Expenditure (CapEx) to Operational Expenditure (OpEx) aligns with CVS's non-profit financial model, converting large upfront investments into predictable monthly costs (Ruparelia, 2016).
Weaknesses	Migration of 6.4GB of critical donor records carries inherent risk of data loss or corruption during transition. There could be internet connectivity problem across all 260 shops, also which may be inconsistent in some locations. The staff need to be retrained across the organisation, and the allocated budget of 10 million must cover implementation, migration, and training costs over three years.
Opportunities	The annual donation income of 96 million can be increased by doing the

	targeted fundraising campaigns through advanced data analytics. The public profile of CVS could also be boosted through the social media integration and modern marketing tools. The business processes can be standardise across all regional offices.
Threats	The cloud security risk such as data breaches, account hijacking, and insecure APIs require thoughtfull mitigation, the data related to donors and beneficiary are sensitive. Also vendor lock-in could lead to disruption in future flexibility. One of the major challenge for the 260 shops is logistics which requires significant planning and resource allocation

2. Analysis of SaaS, PaaS, and IaaS Cloud Solutions for CVS

Cloud computing service models operate at different levels of abstraction, each addressing specific organisational needs (NIST, 2011). All three core models provides the most comprehensive solution for CVS.

2.1 Software as a Service (SaaS)

SaaS is a application delivery mechanism via the internet on a subscription basis which eliminates the need for local installation and maintenance. For CVS, SaaS represents the most impactful service model, directly addressing the organisation's most pressing IT challenges.

CRM Replacement: CVS's 120 marketing and fundraising staff currently depend on failing legacy CRM systems costing £7 million annually. The donor management, campaign tracking, and analytics capabilities can be provided through modern SaaS CRM such as Microsoft Dynamics 365 or Salesforce. The non-profits organisation get offers of 10 free licenses through its power of US programme, along with additional users at 48/user/month (Salesforce.org, 2024). On the other hand, the Dynamics 365 offers upto 75% discount to non-profit organisation with licences from 20/user/month

(Microsoft for Nonprofits, 2024). The 1500 lost working days caused by legacy system are directly get addressed by this system.

HRM and Payroll Consolidation: The current use of five different payroll packages across 16 regional offices is highly inefficient. The processes of payroll can be standardise by using unified SaaS payroll solution such as Sage or Moorepay (approximately £2–4/employee/month; Moorepay, 2025) which would reduce errors, and eliminate the overhead of maintaining multiple system. This would lead to significant reduction in annual expenditure to 96,000-192,000.

Office Productivity: The non-profits organisation get free Business Basic licenses for upto 300 users from Microsoft 365. Also the cost would be 4.40/user/month for Business Premium licenses (Microsoft for Nonprofits, 2024), by replacing the exsiting locally installed office software across all locations.

2.2 Platform as a Service (PaaS)

PaaS provides a development and deployment environment without the complexity of managing underlying infrastructure (NIST, 2011). CVS's six programmers and database support staff can leverage PaaS to build and deploy custom applications efficiently.

E-Commerce Portal Development: The Head of Finance's proposed national e-commerce strategy requires a bespoke platform where 260 shops can upload photographs of valuable items for central pricing and sale. The scalable hosting environment for the e-commerce portal is provided by Azure App Service or AWS Elastic Beanstalk which enable rapid development without the need of provisioning servers. The platform like Shopify (from £25/month; Shopify, 2025) could accelerate the delivery when integrated.

Custom Reporting and Analytics: PaaS enables the development of tailored reporting dashboards to replace the fragmented Access databases and Excel spreadsheets used across regional offices, providing real-time operational and strategic management data.

The fragmented Access databases and Excel spreadsheets can be replaced by enabling PaaS which can help get the tailored reporting which then can be used across offices providing the real-time operational and management data.

2.3 Infrastructure as a Service (IaaS)

Virtualised computing resources such as servers, storage, and networking can be provided by IaaS. For CVS, IaaS forms the foundation layer supporting the migration from on-premises infrastructure.

Compute and Storage: CVS's 6.4GB of critical donor and member data can be hosted on cloud virtual machines and storage services. Azure Virtual Machines (e.g., D2s v5 at approximately £35–40/month with 1-year reserved pricing; Azure Pricing Calculator, 2026) provide reliable, scalable compute capacity, while Azure Blob Storage at approximately £0.015/GB/month (Azure Pricing Calculator, 2026) offers cost-effective data storage with built-in redundancy and disaster recovery.

Network Infrastructure: IaaS eliminates the £6.4 million currently spent on hardware and network purchase and service costs, replacing physical infrastructure with virtual networks connecting head office, 16 regional offices, and 260 shops through secure cloud networking.

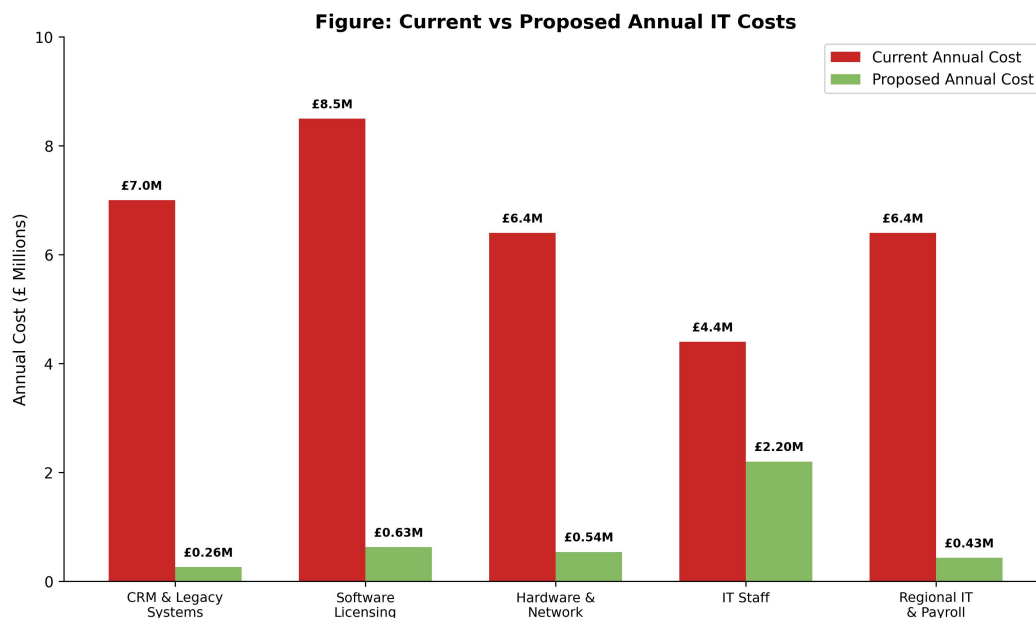


Figure 4: Current vs Proposed Annual IT Costs

3. Service Provider Recommendations and Cost Estimates

Following analysis of major cloud providers, a Microsoft Azure-centric solution is recommended for CVS, supplemented by specialist SaaS services where appropriate. This recommendation is informed by the Peterborough City Council case study, which achieved 40–60% infrastructure cost savings through cloud migration (AWS, 2019), and Microsoft’s industry-leading nonprofit programme.

3.1 Rationale for Microsoft Azure

Microsoft offers the most comprehensive nonprofit programme among major cloud providers, including £2,800/year in Azure credits, free Microsoft 365 licenses, and up to 75% discount on Dynamics 365 (Microsoft for Nonprofits, 2024). Azure’s UK South and UK West data centres ensure data residency compliance, and its enterprise integration with Active Directory and Office tools aligns with CVS’s existing workflows.

3.2 Three-Year Cost Estimate

Service	Annual Cost (Est.)	3-Year Total	Source
Microsoft 365 (4,000 users, nonprofit pricing)	£211,200	£633,600	Microsoft for Nonprofits (2024)
Dynamics 365 CRM (120 users, nonprofit)	£86,400	£259,200	Microsoft for Nonprofits (2024)
Payroll SaaS (4,000 employees)	£144,000	£432,000	Moorepay (2025)
HRM SaaS	£192,000	£576,000	SAP SuccessFactors (2025)
Azure IaaS (VMs, storage, networking)	£180,000	£540,000	Azure Pricing Calculator (2026)
Azure PaaS (e-commerce portal)	£36,000	£108,000	Azure Pricing Calculator (2026)
Migration and	—	£2,000,000	Industry estimate

implementation costs			(Gartner, 2023)
Staff training and change management	—	£500,000	Industry estimate (Gartner, 2023)
Total estimated 3-year cost		£5,048,800	

The proposed solution falls well within the £10 million budget while delivering estimated annual IT cost savings of approximately £18 million compared to the current £25.7 million annual IT expenditure. Even accounting for the retained IT staff costs, net savings over three years are substantial, and funds can be redirected to CVS’s core charitable services.

Figure: Estimated 3-Year Cloud Migration Cost Distribution
Total: £5,048,800

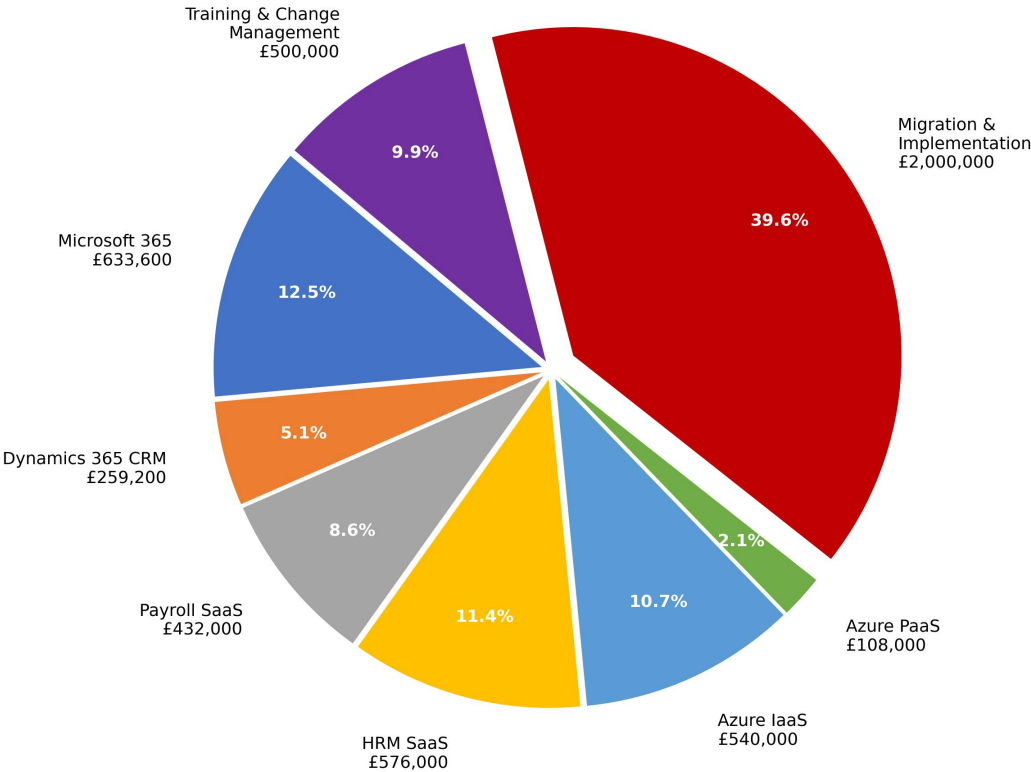


Figure 5: Estimated 3-Year Cloud Migration Cost Distribution

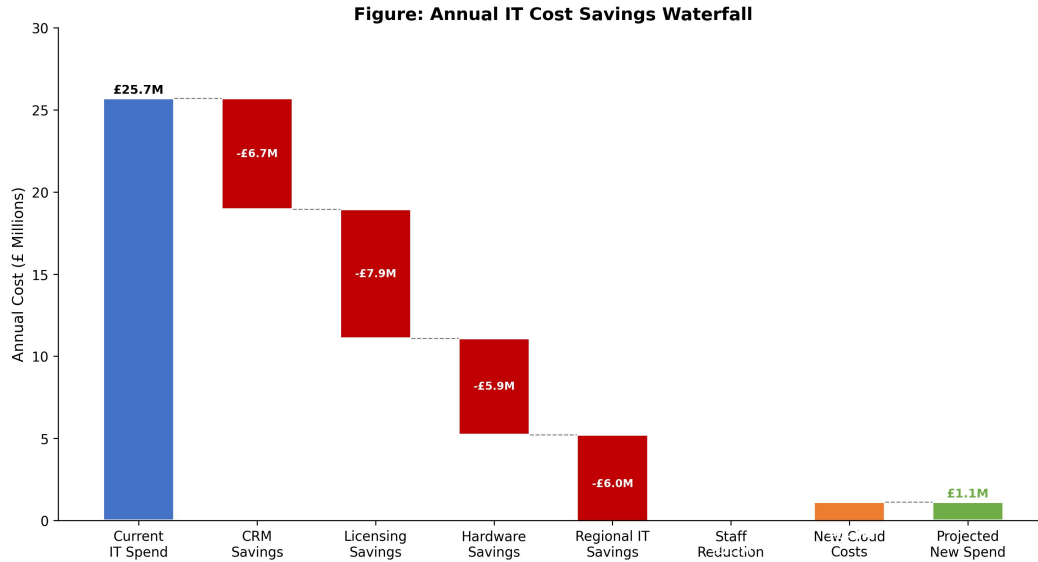


Figure 6: Annual IT Cost Savings Waterfall

3.3 Alternative: AWS

AWS offers strong IaaS capabilities and the Imagine Grant programme provides cloud credits of up to \$100,000 to qualifying nonprofits (AWS, 2024). However, AWS lacks Microsoft’s breadth of nonprofit SaaS discounts, making it less cost-effective as a primary provider for CVS’s needs, though it could serve as a secondary provider in a multi-cloud strategy for redundancy.

4. INaaS and BPaaS Development Recommendations

4.1 Information as a Service (INaaS)

INaaS encompasses database management, content management, information integration, and analytics delivered as cloud services (Haque, 2020). For CVS, INaaS development should follow a phased three-year roadmap:

Year 1 – Data Migration and Integration: Merge all the 6.4GB data of donor and members from legacy system to a unified cloud infrastructure using Azure SQL Database. To eliminate the inconsistencies across all the legacy system, implement data cleansing and master data management.

Year 2 – Analytics and Reporting: The real-time strategic and operational management analytics can be done by deploying Power BI dashboards which will enable the marketing team to conduct data driven analysis, also leading to increase in the annual income from 65 million.

Year 3 – Advanced Analytics: For donor engagement and fundraising optimisation, the predictive analytics can be implement. Also to ensure all the GDPR compliance and information, establish proper data governace frameworks.

4.2 Business Process as a Service (BPaaS)

BPaaS delivers business processes over the internet, shifting from CapEx to OpEx while standardising operations (IBM, 2023). For CVS, BPaaS addresses the fragmentation across 16 regional offices:

Year 1: Standardise payroll and HR processes across all offices through a single BPaaS provider, eliminating the five disparate payroll packages. Implement volunteer management workflows accessible from all 260 shops.

Year 2: Launch the e-commerce fulfilment process as a service, addressing the outbound logistics challenge by partnering with fulfilment BPaaS providers for packaging, insurance, and shipping from individual shops.

Year 3: Full process optimisation including automated donation processing, integrated financial reporting, and streamlined shop management workflows, enabling CVS to operate as a truly unified national organisation.

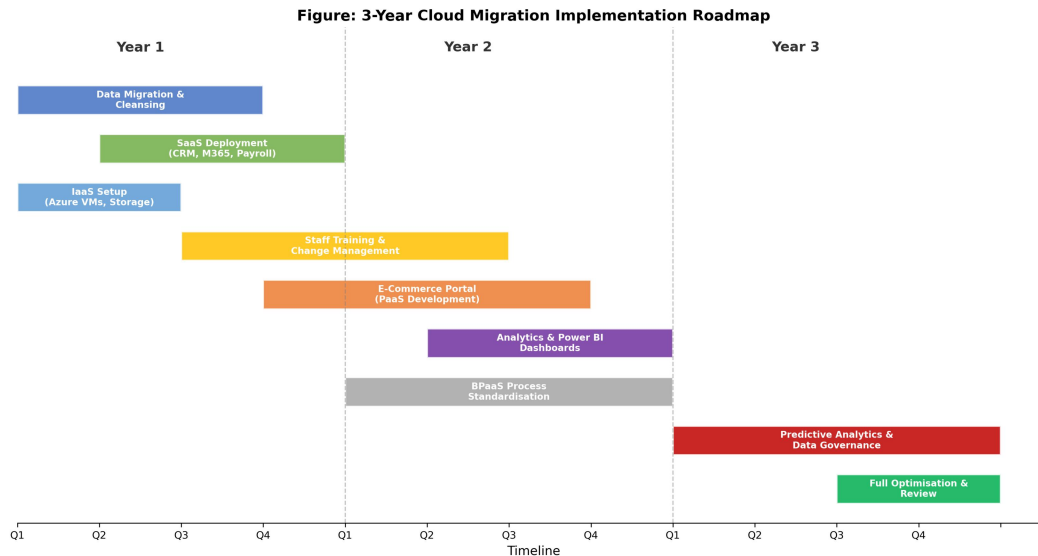


Figure 7: 3-Year Cloud Migration Implementation Roadmap

5. Sustainability and Green Credentials

The value driven charity organisation such as CVS considers the environmental impact carefully. Migrating to cloud data center from on-premises infrastructure helps in delivering the sustainable benefits.

Many cloud provider such as Microsoft Azure have committed to reduce the carbon footprint by 2030 to negative and they have also made a declaration to remove all the historical carbon emmissions by 2050 (Microsoft Environmental Sustainability Report, 2024). They also have proposed to run all the data centres to run on 100% renewable energy.

When comparing microsoft with Google Cloud, it seems they have achieve industry-leading PUE of approximately 1.10 and Google cloud has become carbon neutral since 2007 (Google Environmental Report, 2024). On the other hand, the co-founder of the Climate Pledge of AWS has made a target to become net-zero carbon by 2040 (Amazon Sustainability Report, 2023).

For CVS specifically, migrating from locally managed servers across head office and 16 regional offices to shared cloud infrastructure consolidates energy consumption into purpose-built, highly efficient facilities. There have been many research which indicates the cloud migration can reduce energy consumption by upto 65% when compared to on-premises due to economics of scale optimised cooling, and higher server utilisation rates (Masanet et al., 2020).

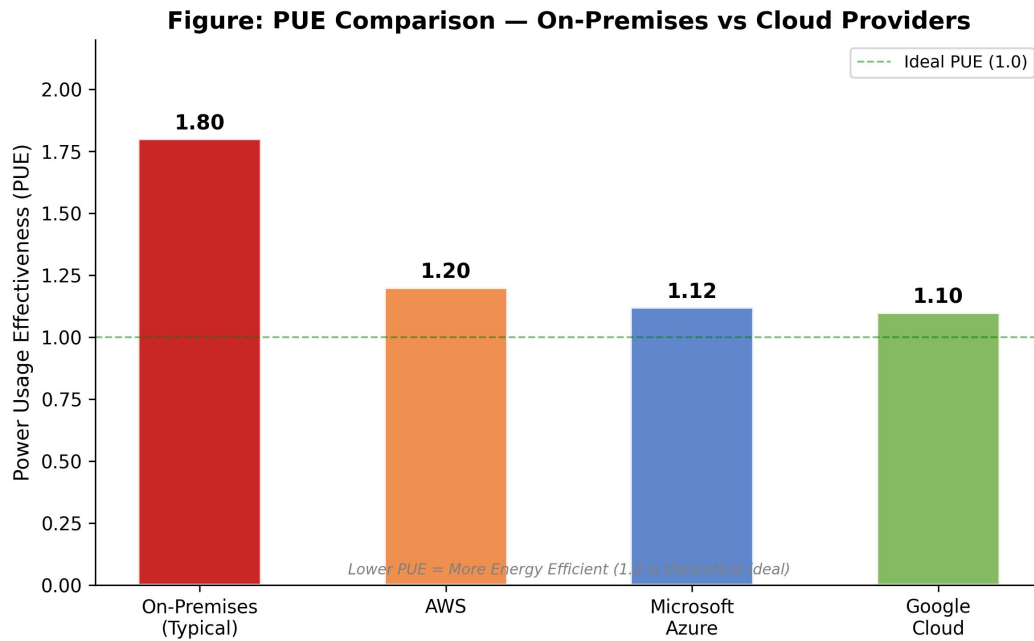


Figure 8: PUE Comparison - On-Premises vs Cloud Providers

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